

Francis Eze

Software Engineer(Fullstack)

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Profile

A detail-oriented software engineer with a solid foundation in backend development, API design, and systems integration. Proficient in Python, Django, FastAPI, Go, and JavaScript frameworks, with experience deploying scalable web applications using Docker, DigitalOcean, and CI/CD tools. Demonstrated ability to solve business problems through code, with recent projects spanning payments, industrial predictive maintenance, e-commerce, healthcare automation, cybersecurity, and financial decision engines. Skilled at cross-functional collaboration and technical writing, with a passion for building useful, performant, user-centric software.

Skills

Languages & Frameworks



Python, Django, Django Rest Framework, JavaScript, Node.js, Express, Docker, React, Next.js, C, Go, Typescript

Databases



MySQL, PostgreSQL, MongoDB, Prisma, Convex

Cloud & DevOps



AWS, Digital Ocean, Docker, Kubernetes, GitHub Actions, and GCP

Professional Experience

Software Engineer/Web Administrator, EPINEC NG

12/2022 – 07/2025

- Revamped and optimized the existing codebase of the company's website, eliminating 80+ bugs and reducing page load time by 40%, resulting in a 25% increase in website conversions. **Increased traffic and feedback by 60%.**
- Developed the frontend and backend architecture for shieldforce.io, resulting in a **40% increase** in website performance and a **25% decrease** in page load time, optimizing the user experience.
- **Worked on third-party API integrations for shieldforce.io.** Responsible for the deployment and maintenance of the website.
- Built the frontend and backend for epinec.us. Added .csv format export feature and linked backend to HubSpot via webhook for lead creation.
- **Led the launch of shieldforce.io from the SE department.**

Inhouse Developer, Kafz Luxury Apartments

08/2022 – 12/2022

- helped to maintain more technical aspects of the company's digital presence.
- **increased the company's online presence by 100%** by creating a landing page, hosting it and also placing the company on Google Maps and setting up the Google Business account.

Projects

Predictive Maintenance Platform , <i>ML-powered platform for industrial equipment failure prediction and maintenance scheduling</i>	2025 – 2026
• Built a FastAPI backend with a scikit-learn prediction service that scores equipment failure risk from live sensor readings (vibration, temperature), enabling proactive rather than reactive maintenance. • Designed asset, sensor, prediction, maintenance log, and alert data models using PostgreSQL and SQLAlchemy, enabling simultaneous monitoring of multiple equipment assets. • Implemented a threshold-based alerting service that flags anomalous sensor readings by severity while preventing duplicate active alerts per asset. • Built a background job scheduler (APScheduler) to run periodic sensor evaluation and prediction jobs automatically, without manual triggering. • Developed a React (Vite) dashboard with Recharts visualizations and Zustand state management, giving operations teams real-time visibility into asset health, active alerts, and maintenance reporting/analytics. • Secured the platform with JWT-based authentication and Argon2 password hashing. • Containerized backend and frontend as independent Docker services for consistent, repeatable deployment.	
Payment Gateway  , <i>Secure transaction processing platform</i>	2025 – 2025
• Designed and implemented a secure payment gateway service to handle end-to-end transaction processing, including payment initiation, verification, and status reconciliation. • Built robust API endpoints for handling card and bank-based payments, ensuring idempotency and consistency across retries. • Implemented webhook handling and signature verification to securely process asynchronous payment events, reducing failed or duplicated transactions by 90%. • Added transaction logging, auditing, and reconciliation logic, improving financial traceability and reducing manual verification effort by 70%. • Integrated fraud checks and basic risk validation rules, preventing invalid or suspicious transactions before processing. • Containerized the service using Docker and deployed it with environment-based configurations, improving deployment reliability and scalability.	
Mini Loan Decision Engine , <i>A lightweight financial decision engine for automating loan approvals based on risk analysis.</i>	2025 – Present
• Designed and implemented a modular rule-based engine that evaluates loan applications by analyzing applicant data and creditworthiness criteria. • Incorporated score computation logic using weights across predefined loan eligibility parameters. • Wrote comprehensive unit tests using Jest to ensure reliable and predictable loan decisions. • Employed TypeScript and Node.js to build a maintainable and extensible decision service, later integrated into a larger loan origination system.	
District Vibes  , <i>Event ticketing and sales platform</i>	2025 – 2025
• Built a full-stack ticketing platform for a live events brand, enabling event creation, ticket sales, and attendee management through a centralized dashboard. • Designed and implemented RESTful APIs for event listings, ticket inventory, and order processing, supporting concurrent ticket purchases without overselling.	

- Integrated payment processing to handle ticket purchases, resulting in **100% automated sales flow** and eliminating manual payment confirmation.
- Implemented QR-code-based ticket generation and validation, reducing event check-in time by **60%** and preventing duplicate ticket usage.
- Optimized database queries and API response times, achieving a **35% reduction in checkout latency** during peak purchase periods.
- Successfully deployed the platform to production and supported a live event, handling real-world user traffic and transactions without downtime.

RediBuy API , *RediBuy is an e-commerce API built using FastAPI.*

2024 – present

- Incorporated features such as authentication and order management through a shopping cart.
- Implemented coupon management as well as payment processing by integrating PayStack.
- Incorporated a real-time collaborative recommendation engine that caches user activity based on their purchase and browsing history on the store, returning products similar to their purchases to your profile on the store, leading to an increase in transaction rate by **20%**.
- Implemented task queuing by using Redis and Celery to run certain tasks in the background to optimize the application's performance, leading to a **30%** rise in application performance.
- Containerized the API using Docker and deployed it using a DigitalOcean droplet.
- Implemented reverse proxy management with Nginx to serve the API in the browser.

Covers and Spines ,

2024 – present

A bookstore API built with Django and the Django REST Framework

- Incorporated features such as authentication, book uploads, as well as payment processing.
- Deployed the application on Heroku after containerizing it using Docker.
- Implemented clear, efficient documentation after completing the API.

Virtual Doctor, *A web application that mimicks real life hospital checkins*

2024 – 2024

- Collaborated with a team made up of designers and frontend developers to create a web application that could help arrange appointments, receive preliminary diagnoses, and provide access to testing and pharmacy facilities around the user.
- Implemented the backend and created the endpoints to handle booking an appointment and handling patient information.
- Integrated Google's Maps and Places API to create endpoints for laboratory and pharmacy locators.
- Integrated OpenAI's chatgpt3.5 turbo API to create an endpoint that provides a diagnosis of a user's symptoms through a chatbot. To achieve this, I created a symptom checker algorithm that checks if there are symptoms in the user's message, checks them against the list of symptoms and diseases in the database, and returns a possible diagnosis based on the symptoms.

EPINEC LLC Official Website, *Landing Page*

2023 – 2024

- Implemented both the front and backends for epinec.us , an IT consultancy firm and a subsidiary of EPINEC LLC
- Incorporated CSV file format export feature for contact form data for accessibility by spreadsheet applications such as Excel.
- Managed DNS retrieval for the domain name and deployed both the front- and back-end applications using Digital Ocean's App Platform.

ShieldForce 🔗 , <i>A B2B Cybersecurity subscription platform</i>	2023 – 2024
• Implemented both the front and back ends of shieldforce.io, a B2B subscription platform for cybersecurity solutions. • Integrated third-party APIs from providers such as Chargebee, HubSpot, and SendGrid. • Deployed the front- and back-end applications using Digital Ocean's App Platform.	

Education

Electronic and Computer Engineering , <i>Nnamdi Azikiwe University</i>	2019 – 2024
Diploma , <i>ALX Software Engineering Program</i>	2022 – 2023
1 year software engineering program hoping to specialize in backend engineering.	